

Salesforce Developer Bridge Program

Day 3 – Self Learning Track Report

Validation Rules, Flows & Triggers

Project: Placement Management System

Student: Sudharshan Kuppili

PIN: 24PA5A0517

Student Details

Field	Details
Name	Sudharshan Kuppili
Project	Placement Management System
Day	Day 3 – Validation Rules, Flows & Triggers (Automation)
Technology	Salesforce (Flow Builder, Validation Rules, Declarative Automation)

1. Objective

The objective of Day 3 is to understand Salesforce Validation Rules, Flows, and Apex Triggers, select the appropriate automation tool for different business requirements, implement Record-Triggered Flows, create validation rules, and enhance the Placement Management System with Offer Letter automation.

2. Business Scenario

The Placement Cell required the following automation on the Placement Management System:

- Send an email notification to the Placement Officer when a student submits an application.
- Automatically populate Application Date when a new application is created.
- Prevent duplicate / incomplete applications by enforcing mandatory fields.
- Reject applications when the student's CGPA is below the Job's minimum CGPA.
- Automatically create an Offer Letter when an application's Status changes to Selected.

3. Design Challenge

For each requirement, the most suitable declarative automation tool was selected as follows:

Requirement	Chosen Tool	Reason
Reject duplicate applications	Flow	Flow can check existing application records and prevent a duplicate based on the Student and Job combination.
Auto-fill Application Date	Flow	A Record-Triggered (before-save) Flow can populate the Application Date automatically when a new application is created.
Send Email	Flow	Flow can send an email notification automatically when a Placement Application is created.
Reject low CGPA	Validation Rule	A Validation Rule can prevent the record from being saved when the student's CGPA is below the minimum requirement.
Create Offer Letter	Flow	A Record-Triggered Flow can detect when Status changes to Selected and create the related Offer Letter record automatically.

4. Application Date Record-Triggered Flow

Purpose

Automatically populate Application Date when a Placement Application is created.

Implementation

- Object: Placement Application
- Trigger: A record is created
- Flow type: Before-save / Fast Field Updates
- Assignment / Update logic: Set Application Date to the current date (\$Flow.CurrentDate)
- Flow saved, activated, and tested successfully.

Expected Result

When a new Placement Application is saved without an Application Date, Salesforce automatically populates the Application Date.

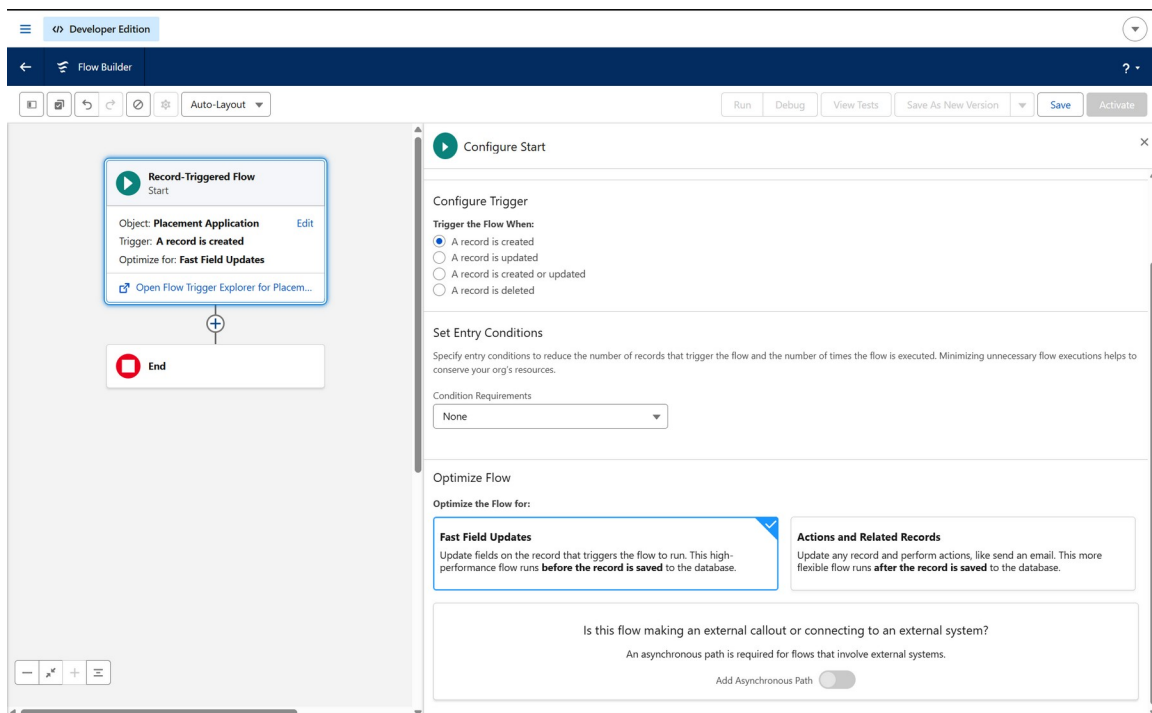


Figure 1: Application Date Flow — Configure Start (Object: Placement Application, Trigger: A record is created, Optimize: Fast Field Updates)

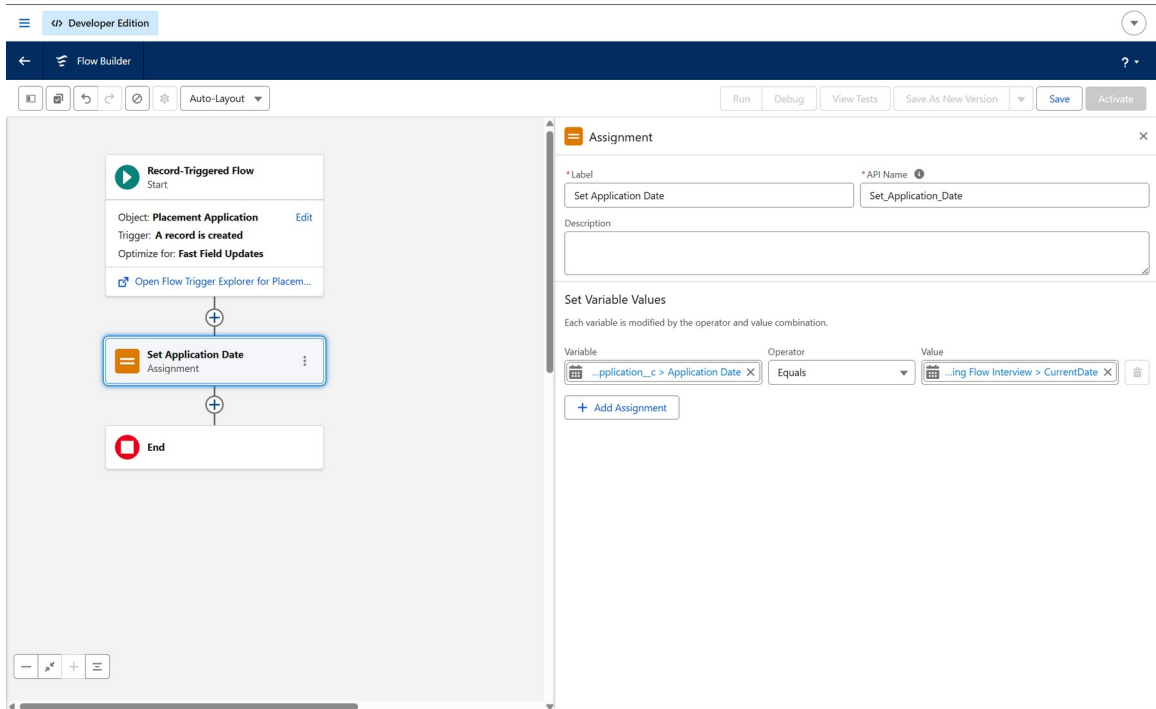


Figure 2: Assignment element — Application Date set to \$Flow.CurrentDate

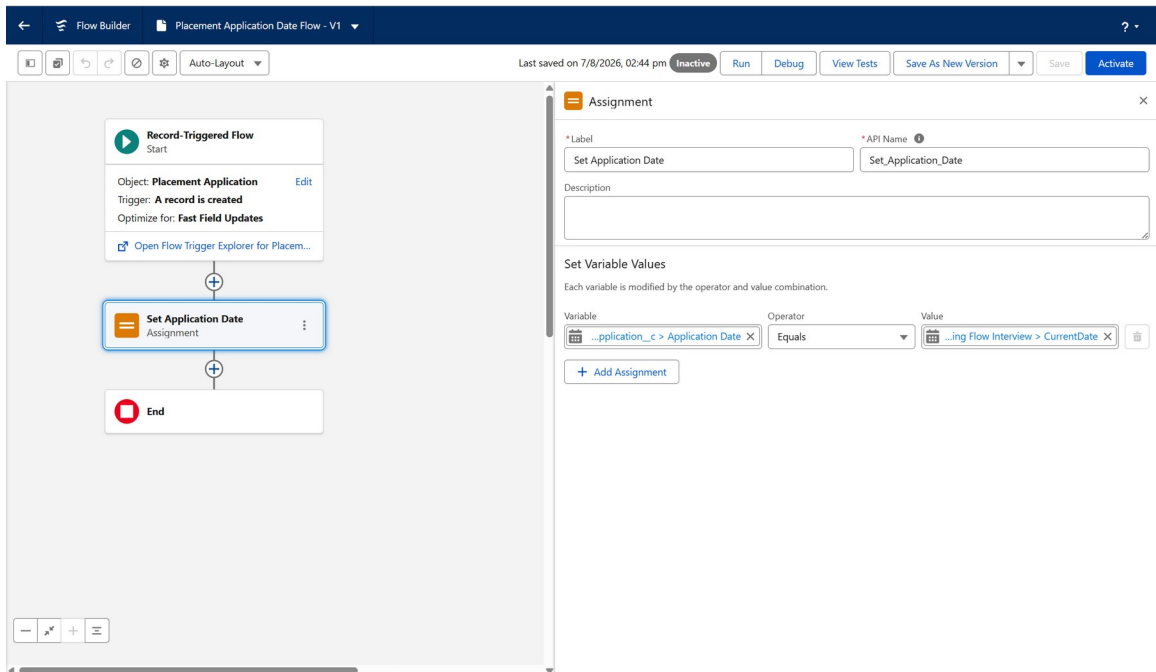


Figure 3: Flow saved as "Placement Application Date Flow - V1" (Inactive, prior to activation)

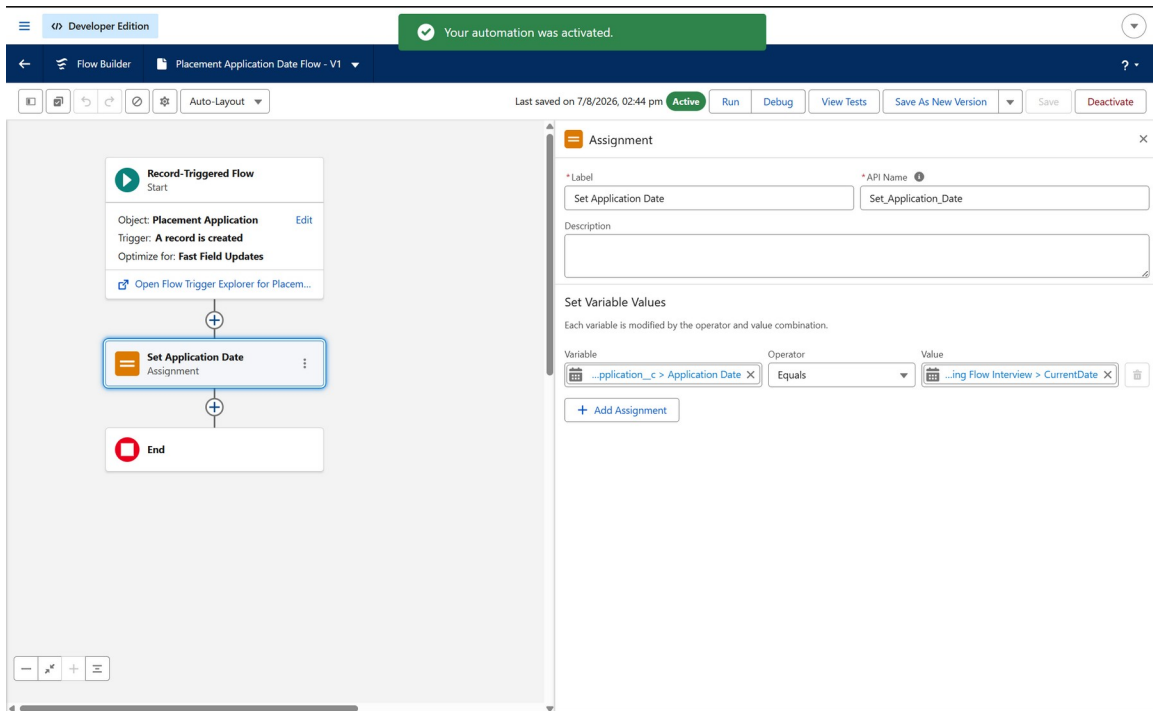


Figure 4: Flow activated — "Your automation was activated" confirmation

5. Email Automation

Purpose

Send a confirmation email whenever a Placement Application is created.

Implementation

- A separate Record-Triggered Flow was created because the Application Date flow is a before-save flow and cannot perform actions such as sending email.
- Object: Placement Application
- Trigger: A record is created
- Optimize for: Actions and Related Records
- Action: Send Email
- Recipient Address: info@vishnu.edu.in
- Sender Type: Current User
- Email content configured using Compose Email Content.

Implementation adaptation: the recipient address used for testing was the organization's general mailbox (info@vishnu.edu.in) rather than a separate "Placement Officer" user record, and the flow was saved/activated as Version 2 (V2) after an initial inactive save (V1) shown below.

Testing

A new Placement Application (PA-0003) was created with Remarks "Email notification test" to trigger the flow.

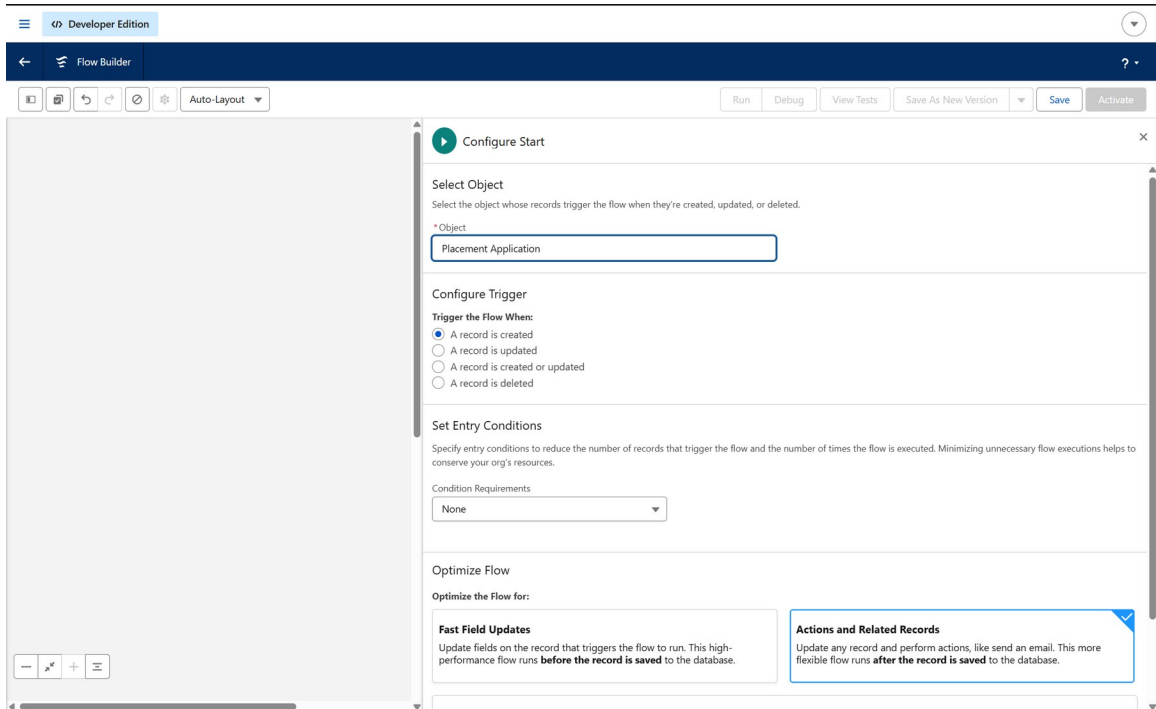


Figure 5: Email Flow — Configure Start (Object: Placement Application, Trigger: A record is created, Optimize: Actions and Related Records)

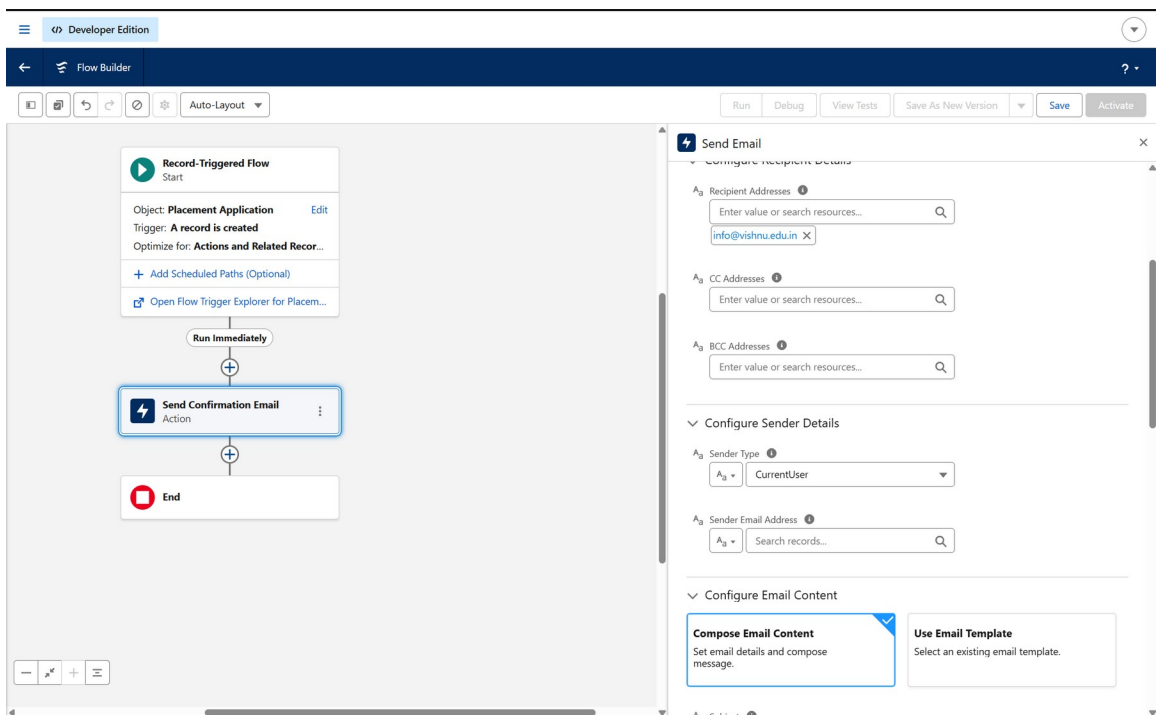


Figure 6: Send Email action — Recipient Address configured as info@vishnu.edu.in, Sender: Current User

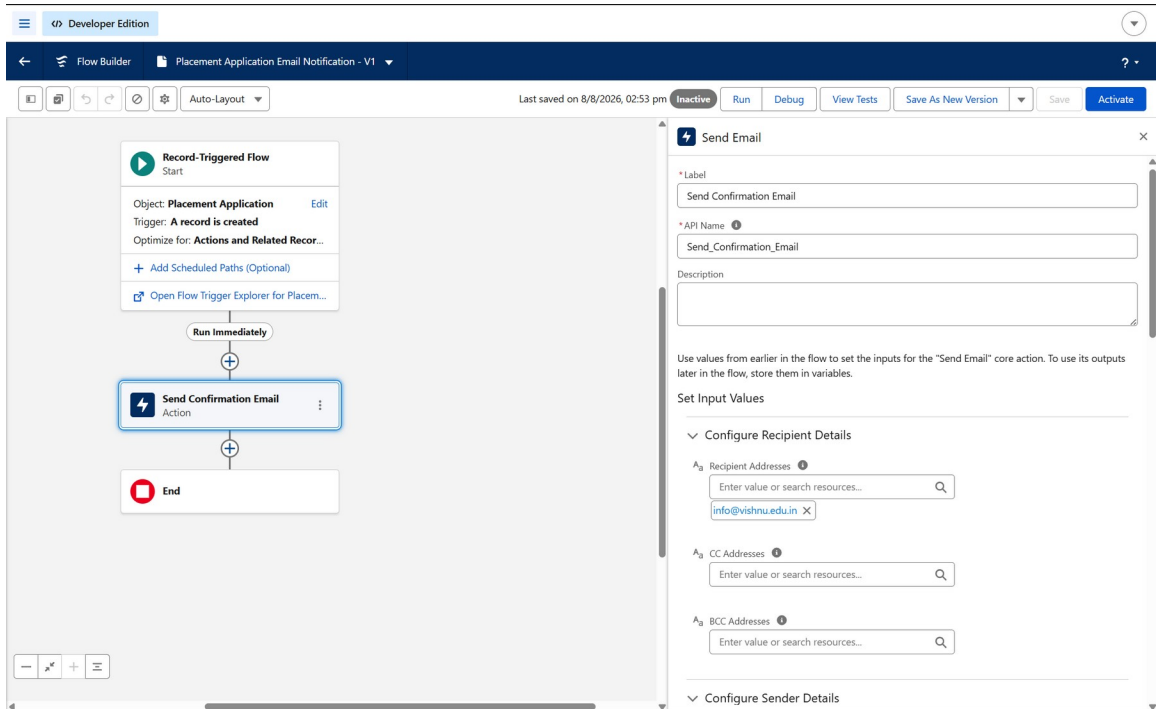


Figure 7: Flow saved as "Placement Application Email Notification - V1" (Inactive) — action labeled "Send Confirmation Email"

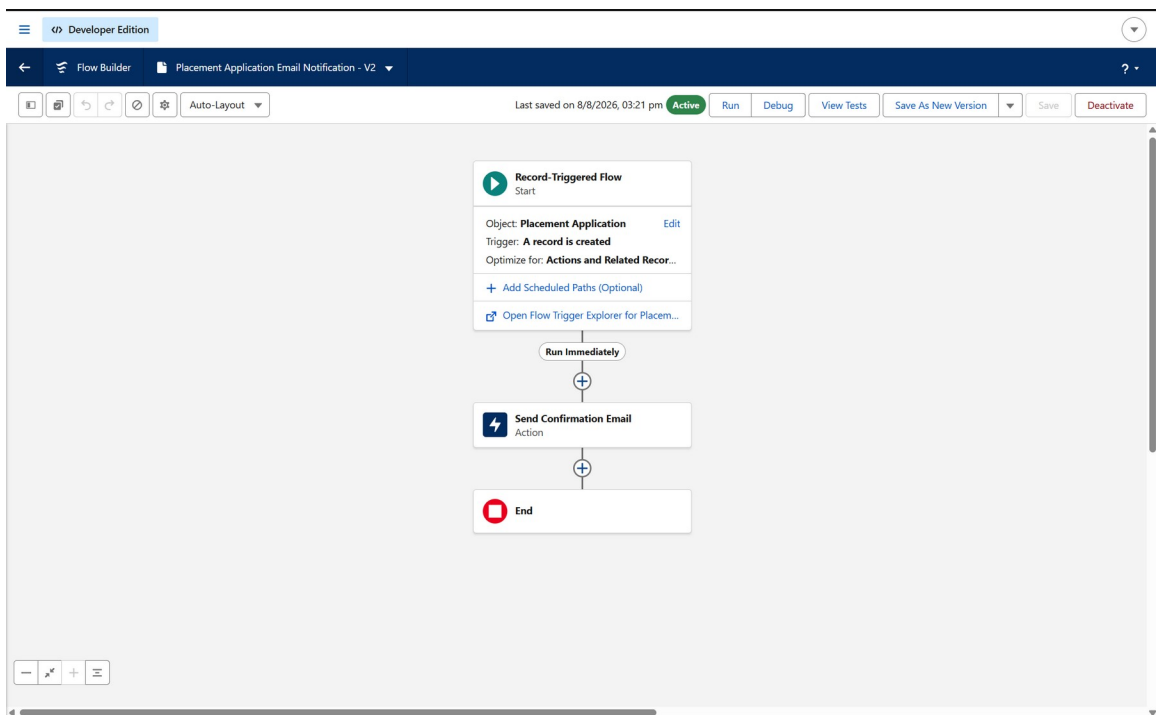


Figure 8: Flow activated as "Placement Application Email Notification - V2" (Active)

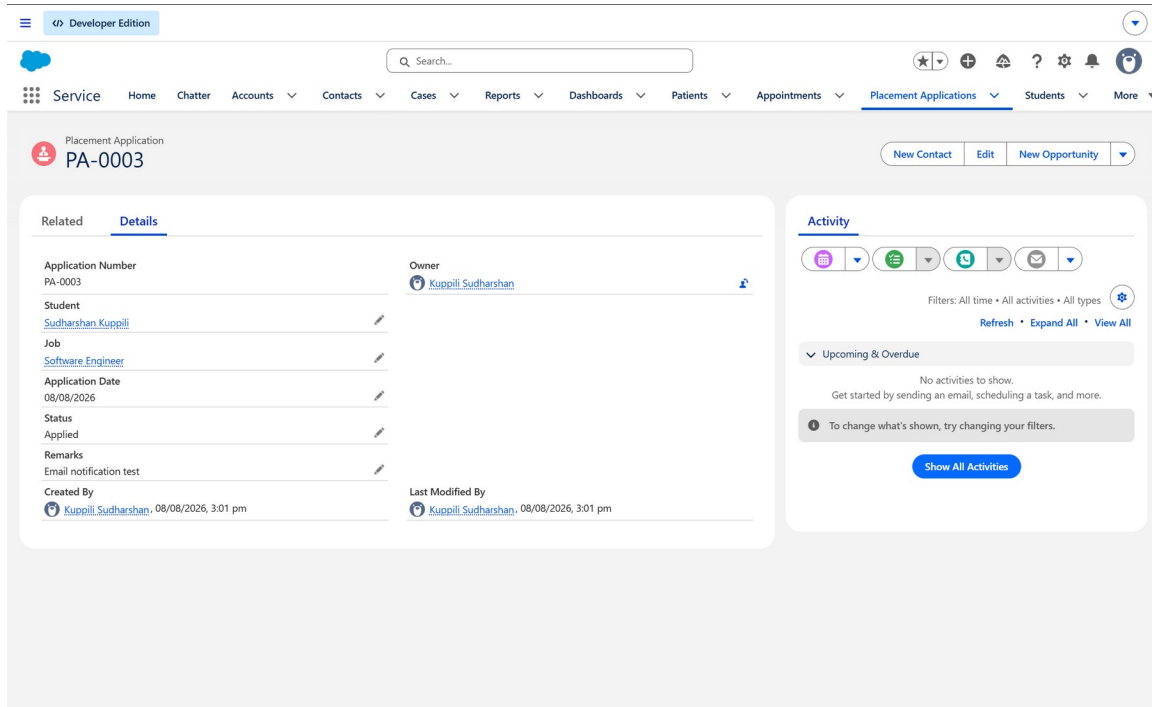


Figure 9: Test record PA-0003 created with Remarks "Email notification test" to trigger the flow

The Send Email action was configured with a fixed recipient address, Current User as the sender, and composed email content. Testing was carried out by creating a new Placement Application record (PA-0003) to trigger the flow; the flow's trigger, action configuration, and activation were verified directly in Flow Builder as shown above.

6. Validation Rules

Five validation rules were implemented on the Placement Application object to enforce data quality and eligibility requirements.

6.1 Student Must Be Selected

Rule Name: Student_Must_Be_Selected

```
ISBLANK(Student__c)
```

Error Message: "Please select a Student before saving the application."

Purpose: Prevents saving a Placement Application without selecting a Student.

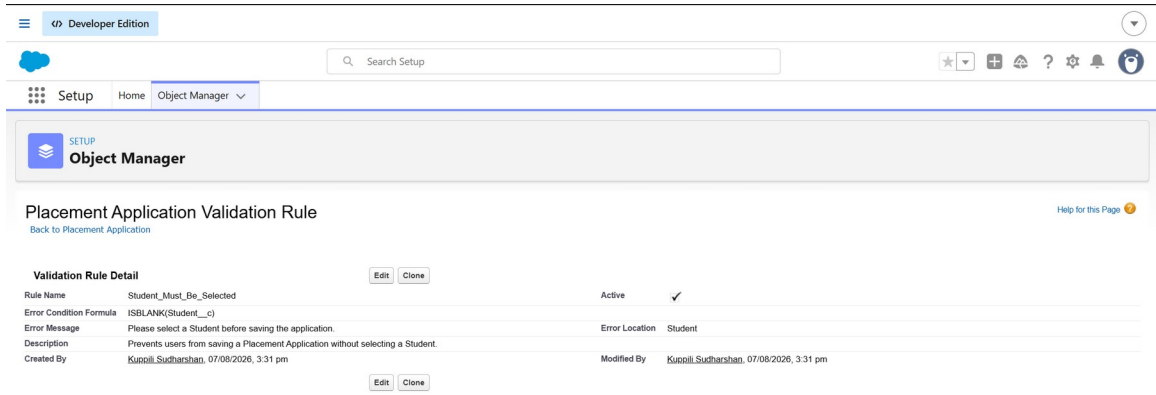


Figure 10: Validation Rule — Student_Must_Be_Selected

6.2 Job Must Be Selected

Rule Name: Job_Must_Be_Selected

ISBLANK(Job__c)

Error Message: "Please select a Job before saving the application."

Purpose: Prevents saving a Placement Application without selecting a Job.

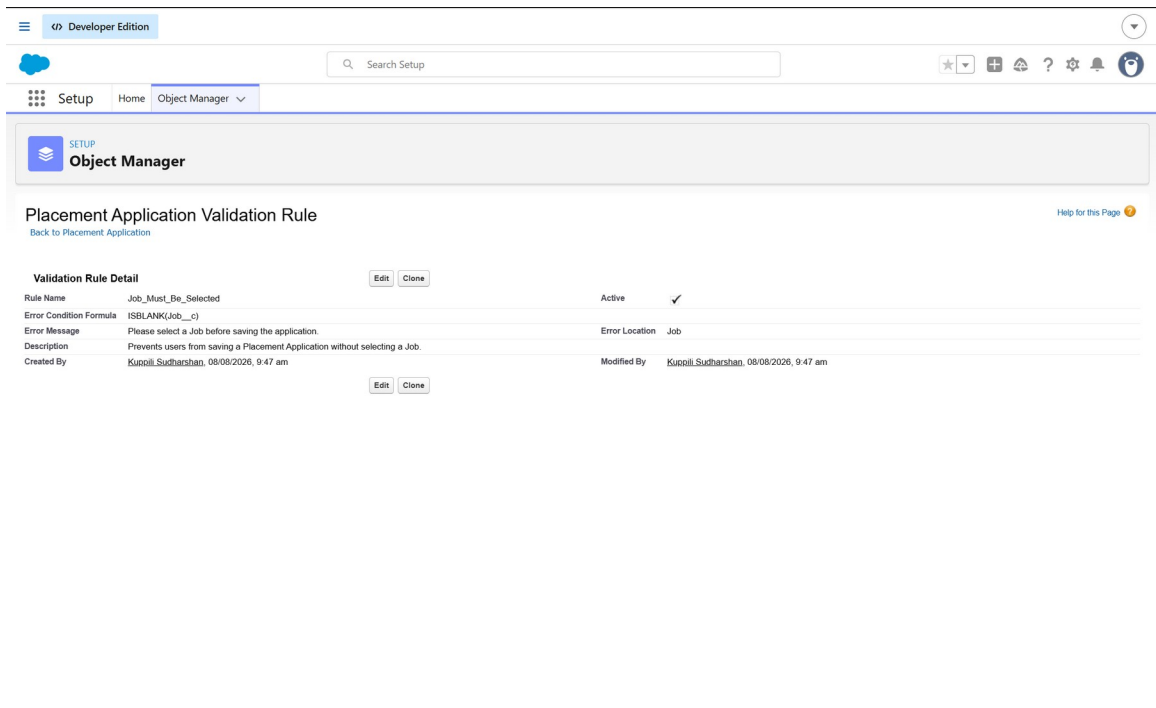


Figure 11: Validation Rule — Job_Must_Be_Selected

6.3 Application Date Must Be Present

Rule Name: Application_Date_Must_Be_Present

ISBLANK(Application_Date__c)

Error Message: "Application Date is required."

Purpose: Prevents a Placement Application from being saved without an Application Date.

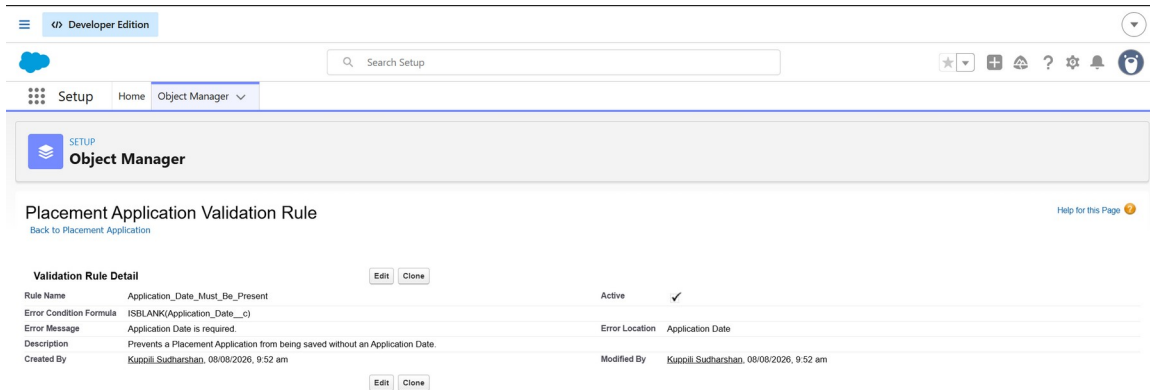


Figure 12: Validation Rule — Application_Date_Must_Be_Present

6.4 Student CGPA Eligibility

Rule Name: Student_CGPA_Eligibility

Student__r.CGPA__c < Job__r.Minimum_CGPA__c

Error Message: "Student CGPA does not meet the minimum CGPA requirement for this job."

Purpose: Prevents a student from applying when the student's CGPA is below the minimum CGPA required for the selected Job.

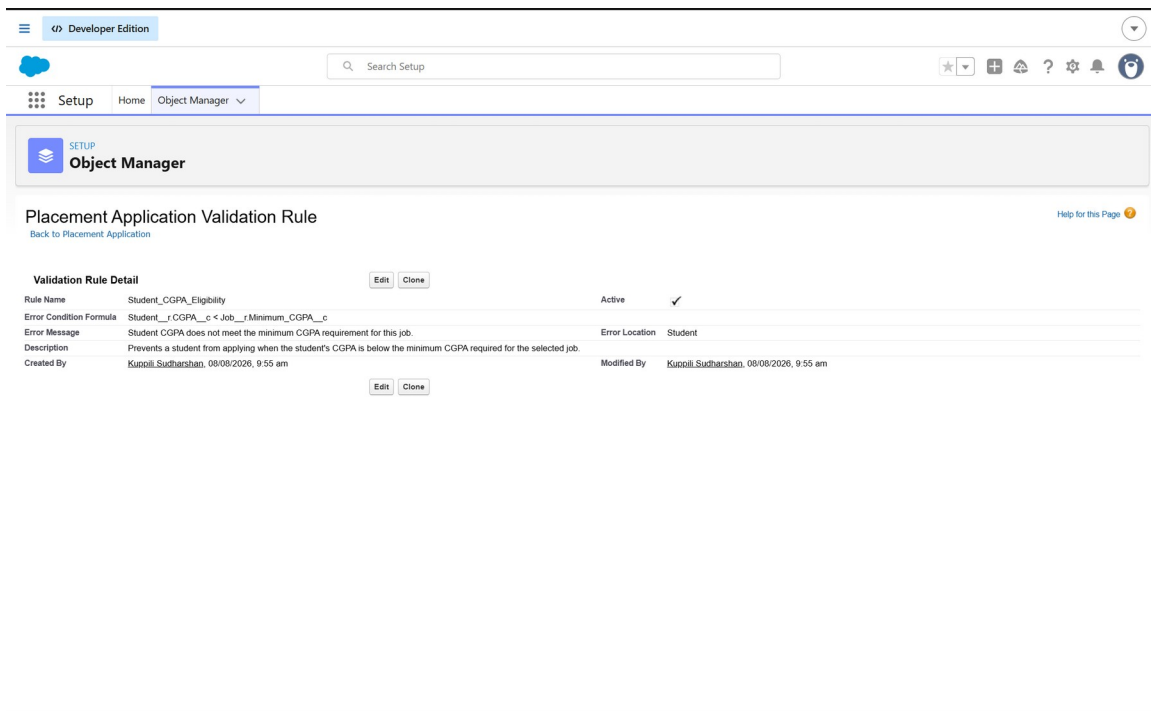


Figure 13: Validation Rule — Student_CGPA_Eligibility

Testing

Valid application: Student CGPA = 8.75 (Sudharshan Kuppili); Job Minimum CGPA = 7.50 (Software Engineer). The application (PA-0002) saved successfully.

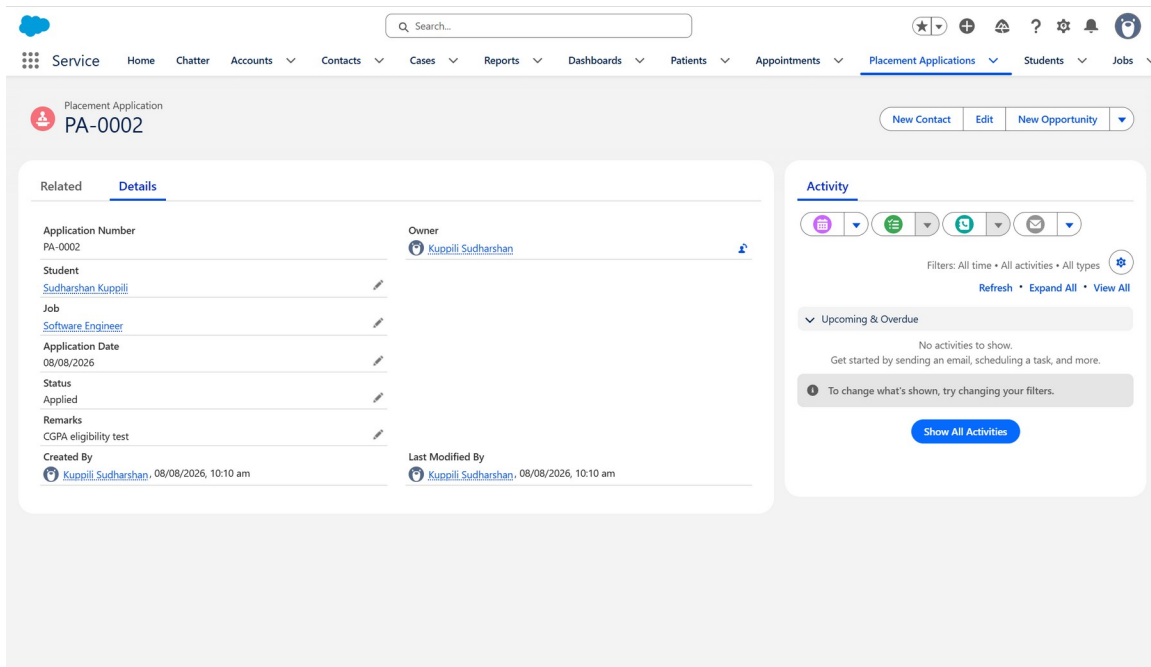


Figure 14: Placement Application PA-0002 saved successfully — CGPA 8.75 meets the Job's minimum of 7.50

Invalid application: a test Job "AI Engineer – Test" was created with Minimum CGPA = 9.00 (above the student's CGPA of 8.75) to demonstrate the rule blocking a save.

Developer Edition

Search...

Service Home Chatter Accounts Contacts Cases Reports Dashboards Patients Appointments Placement Applications Students Jobs

Job AI Engineer - Test

New Contact Edit New Opportunity

Related Details

Job Name AI Engineer - Test

Company Name Test Company

Minimum CGPA 9.00

Eligible Branch CSE

Active Backlog Limit 0

Application Deadline 26/08/2026

Created By Kuppli Sudharshan 08/08/2026, 10:48 am

Owner Kuppli Sudharshan

Last Modified By Kuppli Sudharshan 08/08/2026, 10:48 am

Figure 15: Test Job record "AI Engineer - Test" — Minimum CGPA set to 9.00

Developer Edition

Search...

Service Home Chatter Accounts Contacts Cases Reports Dashboards Patients Appointments Placement Applications Students Jobs

New Placement Application

* = Required Information

Information

Application Number

Student Sudharshan Kuppli

Student CGPA does not meet the minimum CGPA requirement for this job.

Job AI Engineer - Test

Application Date 08/08/2026

Status Applied

Remarks Invalid CGPA test

We hit a snag. Review the following fields: Student

Cancel Save & New Save

Figure 16: Application blocked — "Student CGPA does not meet the minimum CGPA requirement for this job."

6.5 Application Date After Deadline

Rule Name: Application_Date_After_Deadline

`Application_Date__c > Job__r.Application_Deadline__c`

Error Message: "Application cannot be submitted after the job application deadline."

Purpose: Prevents an application from being submitted after the Job's Application Deadline.

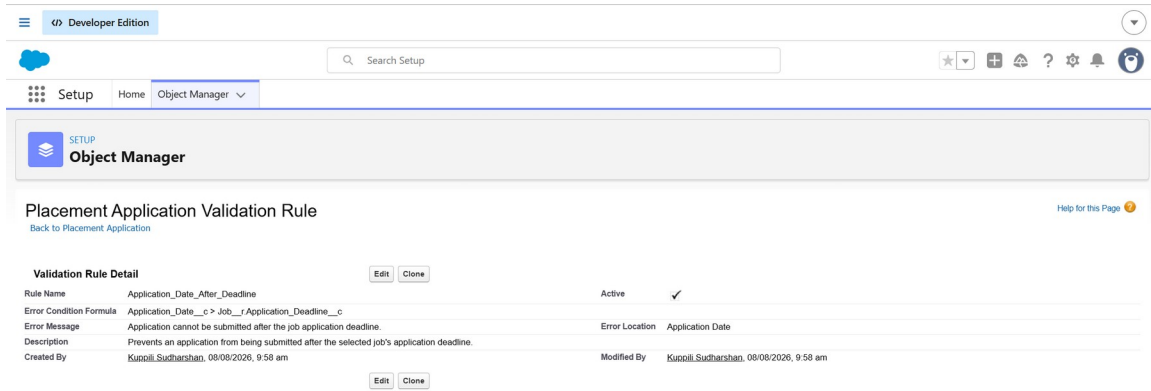


Figure 17: Validation Rule — Application_Date_After_Deadline

7. Flow vs Apex Trigger

Aspect	Flow	Apex Trigger
Nature	Declarative / low-code	Programmatic
Best suited for	Standard business processes, simple-to-medium automation	Complex business requirements, custom logic
Maintainability	Easier to build and maintain	Requires coding, testing, debugging, and ongoing maintenance
Typical use	Field updates, record creation, notifications, most standard automation	Complex multi-object logic, bulk processing, integrations

Interview Summary

Use Flow when declarative automation can solve the requirement. Use Apex when the logic is complex or cannot be handled effectively with Flow. Flow cannot completely replace Apex Triggers — for example, bulk data processing at scale, complex multi-object transactional logic, or callouts requiring fine-grained control are still better suited to Apex.

8. Offer Letter Mini Project Enhancement

Purpose

Automatically create an Offer Letter when a Placement Application's Status changes to Selected.

Offer Letter Object — Offer_Letter__c

The Offer Letter object and its fields were built up incrementally:

The screenshot shows the Salesforce Setup interface in Developer Edition. The left sidebar lists various setup options, with 'Fields & Relationships' selected. The main content area displays the 'Offer Letter' object with 5 fields. The fields are listed in a table with columns: FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Data Type		
Last Modified By	LastModifiedById	Lookup(User)		
Offer Letter Number	Name	Auto Number		✓
Owner	OwnerId	Lookup(User,Group)		✓
Student	Student__c	Lookup(Student)		✓

Figure 18: Offer Letter object — initial fields (Offer Letter Number, Owner, Student)

The screenshot shows the Salesforce Setup interface in Developer Edition, similar to Figure 18, but with an additional field added. The 'Offer Letter' object now has 6 fields. The 'Placement Application' field is a lookup field to the 'Placement Application' object.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD
Created By	CreatedById	Lookup(User)	
Last Modified By	LastModifiedById	Lookup(User)	
Offer Letter Number	Name	Auto Number	
Owner	OwnerId	Lookup(User,Group)	
Placement Application	Placement_Application__c	Lookup(Placement Application)	
Student	Student__c	Lookup(Student)	

Figure 19: Offer Letter object — Placement Application lookup field added

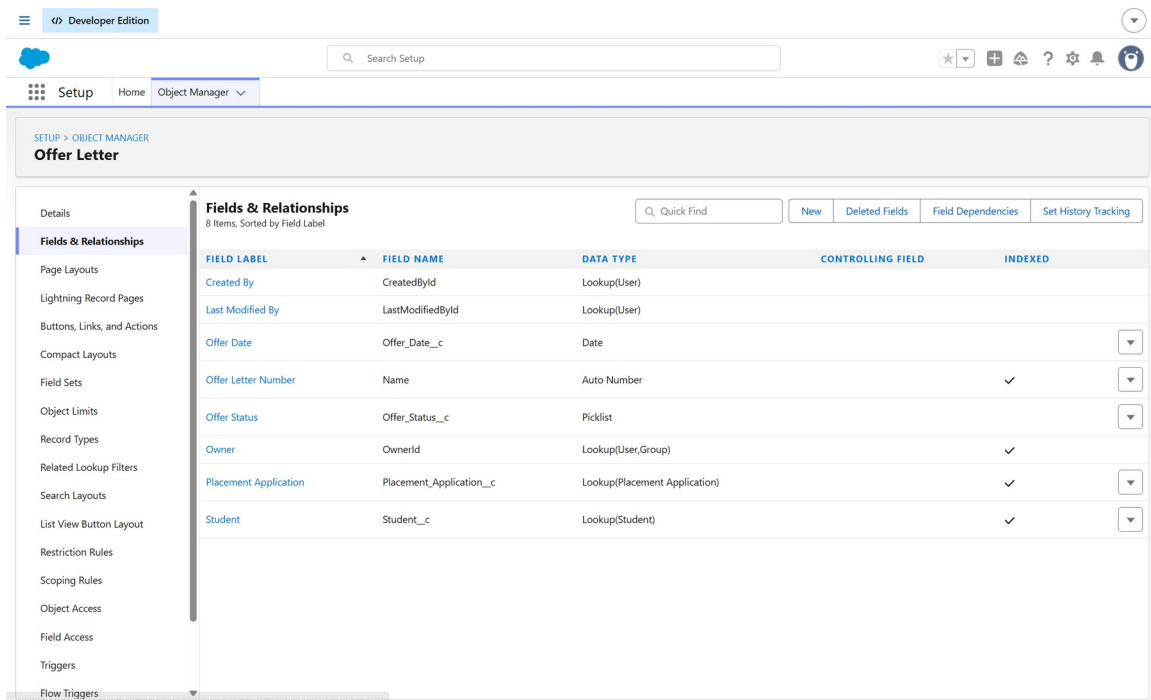


Figure 20: Offer Letter object — final field set, including Offer Date (Date) and Offer Status (Picklist: Generated, Accepted, Rejected)

Flow Configuration

- Object: Placement Application
- Trigger: A record is updated
- Conditions (2): Status Equals "Selected" AND Status Is Changed = True
- Optimize for: Actions and Related Records
- Create Records element: "Create Offer Letter"

Offer Letter field mappings:

```
Student      <- Triggering Placement_Application__c > Student
Placement Application <- Triggering Placement_Application__c > Record ID
Offer Date   <- Running Flow Interview > CurrentDate
Offer Status <- "Generated"
```

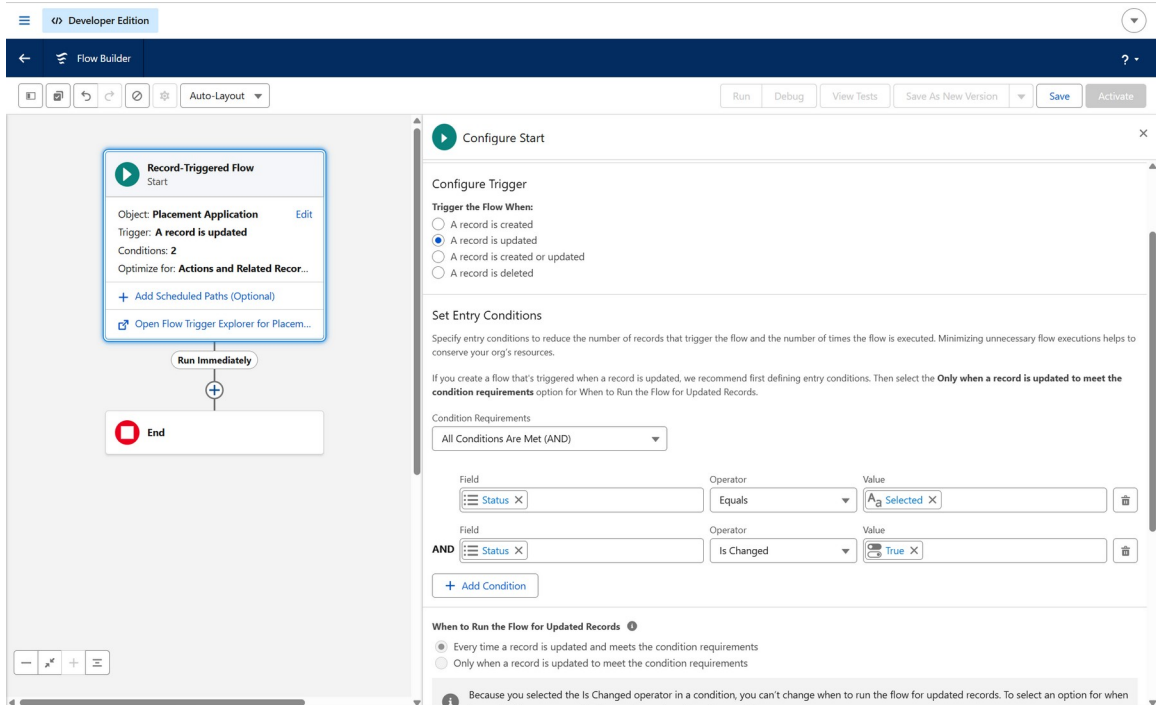


Figure 21: Offer Letter Flow — Configure Start (Trigger: A record is updated, Conditions: 2)

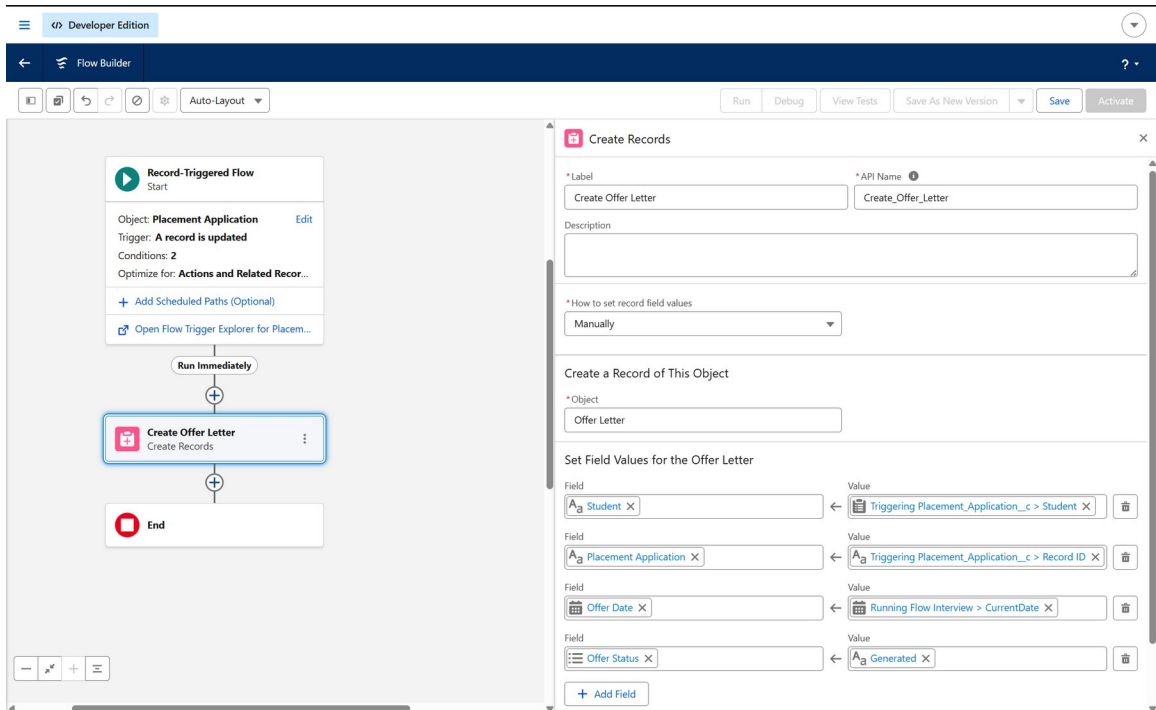


Figure 22: Create Records element — "Create Offer Letter" with field mappings

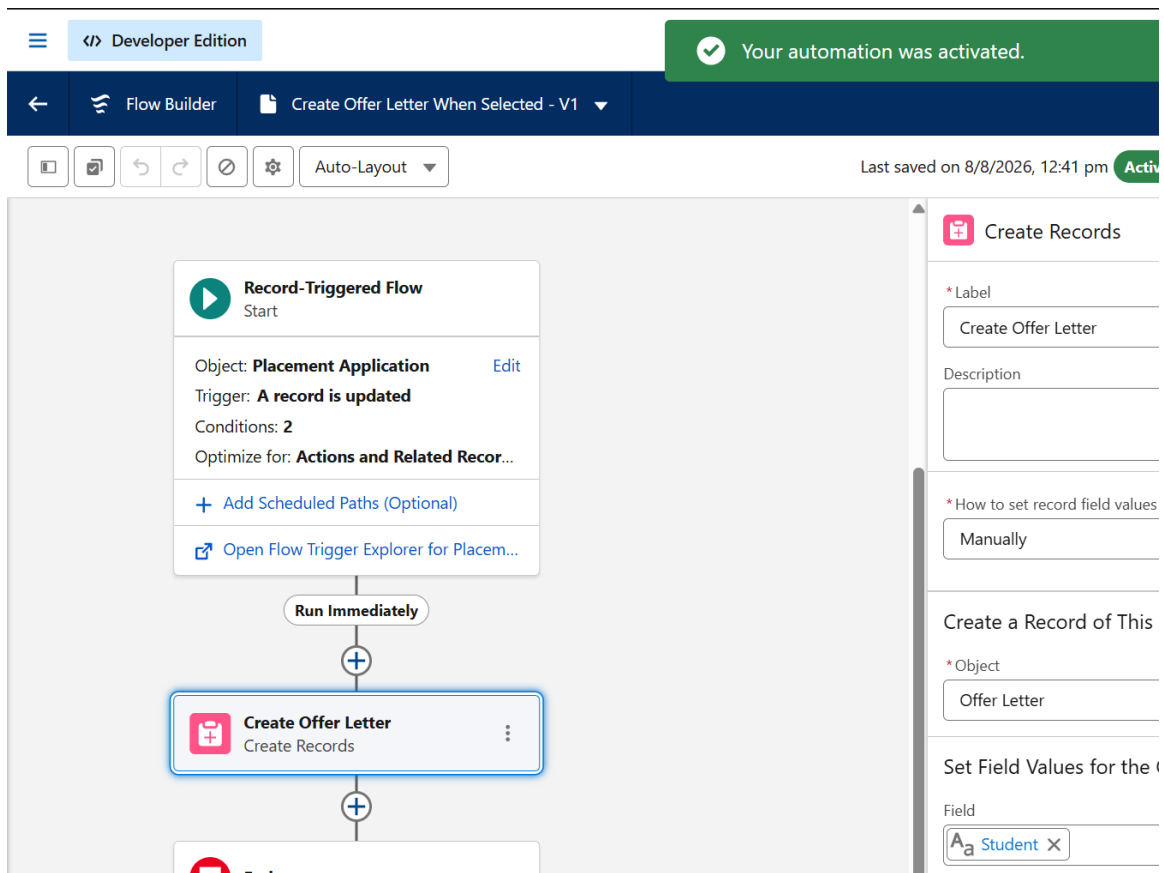


Figure 23: Flow activated — "Create Offer Letter When Selected - V1" (Active)

9. Successful Offer Letter Execution

Test sequence: Placement Application PA-0002 → Status changed to Selected → Flow executed → Offer Letter automatically created.

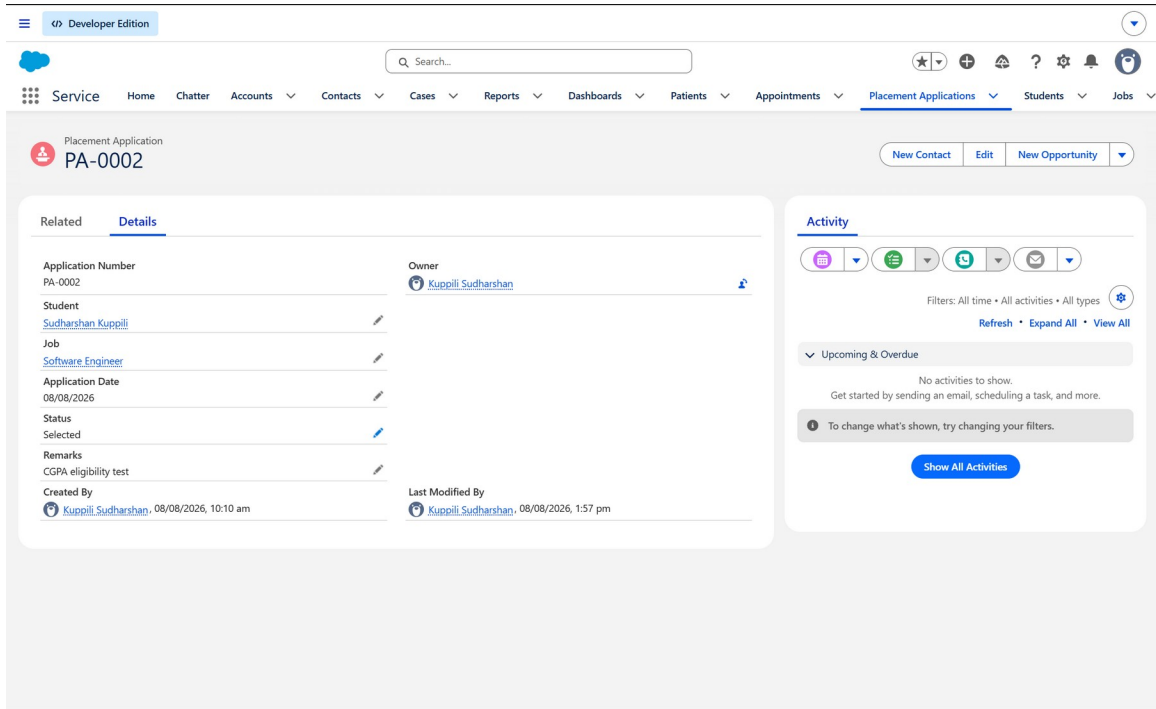


Figure 24: Placement Application PA-0002 with Status changed to "Selected"

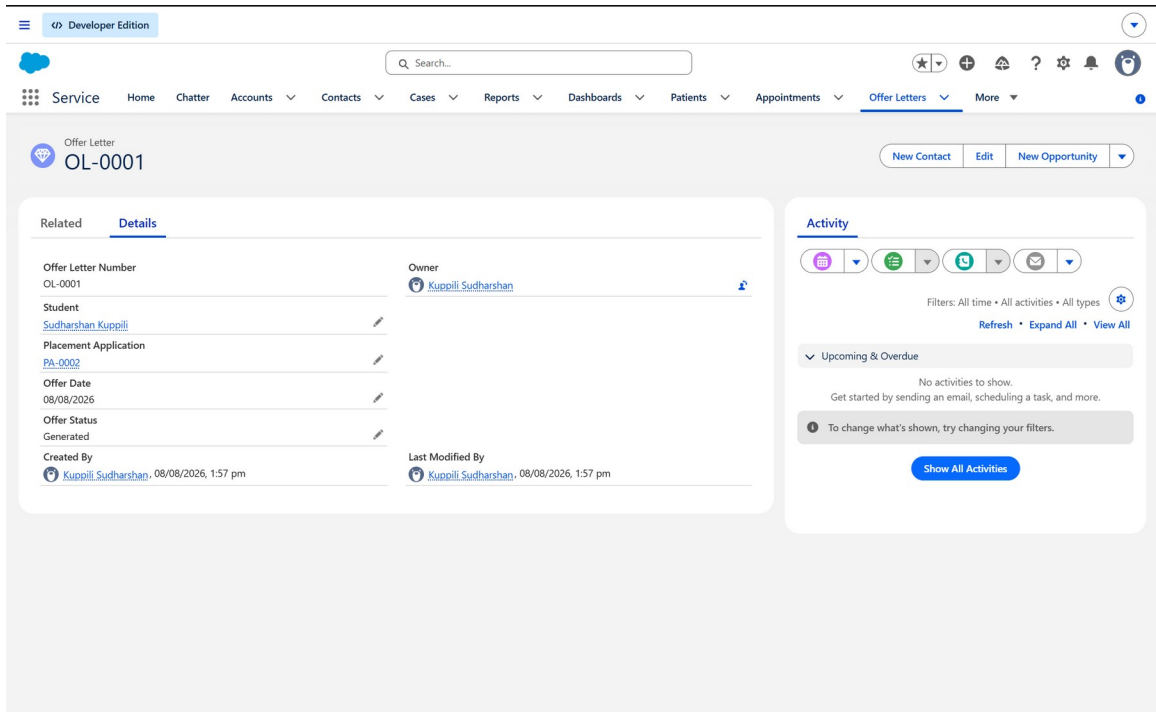


Figure 25: Auto-created Offer Letter record OL-0001 — Student, Placement Application (PA-0002), Offer Date, and Offer Status = Generated all populated automatically

10. Debugging Challenge

Scenario: a Trigger updates Status, a Flow updates Status, and a Workflow Rule also updates Status.

Q1: What problem might occur?

The automations can conflict because multiple automation tools are updating the same Status field. This can cause unexpected results and make the system difficult to debug and maintain.

Q2: Could automation repeatedly execute?

Yes. If one automation updates a field that causes another automation to run again, the automation can repeatedly execute, creating recursion or an automation loop.

Q3: How can it be prevented?

Use the appropriate automation tool for each business requirement and avoid multiple, unnecessary automations updating the same field. Design automation carefully — for example, consolidating logic into a single Flow or Trigger per object — to prevent recursion and conflicting updates.

11. Test Results

Test Case	Expected Result	Actual Result	Status
Application Date auto-population	Application Date automatically populated	Application Date populated	PASS
Confirmation email flow	Email flow triggers and sends confirmation on application creation	Flow activated (V2); test record PA-0003 created to trigger the action	PASS
Valid CGPA application	Application saved	Application PA-0002 saved successfully	PASS
Invalid CGPA application	Application blocked	Validation error displayed and application blocked	PASS
Offer Letter creation	Offer Letter created when Status becomes Selected	Offer Letter OL-0001 automatically created	PASS

12. Key Learnings

During Day 3, I learned how to select the appropriate Salesforce automation tool based on the business requirement. I implemented Record-Triggered Flows, Validation Rules, Email automation, and automatic Offer Letter creation. I also learned why Flow is preferred for declarative automation and why Apex is useful for complex business logic. Finally, I learned how conflicting automation can cause recursion and unexpected results, and the importance of testing both valid and invalid scenarios to confirm validation rules work as intended.

13. Conclusion

Day 3 successfully covered Salesforce Validation Rules, Flows, Apex Trigger concepts, automation design, testing, debugging, and the Offer Letter enhancement for the Placement Management System. All planned automations were implemented, activated, and verified with real screenshots.

Appendix A — Supporting Test Data

The following records were used to test the automations documented above.

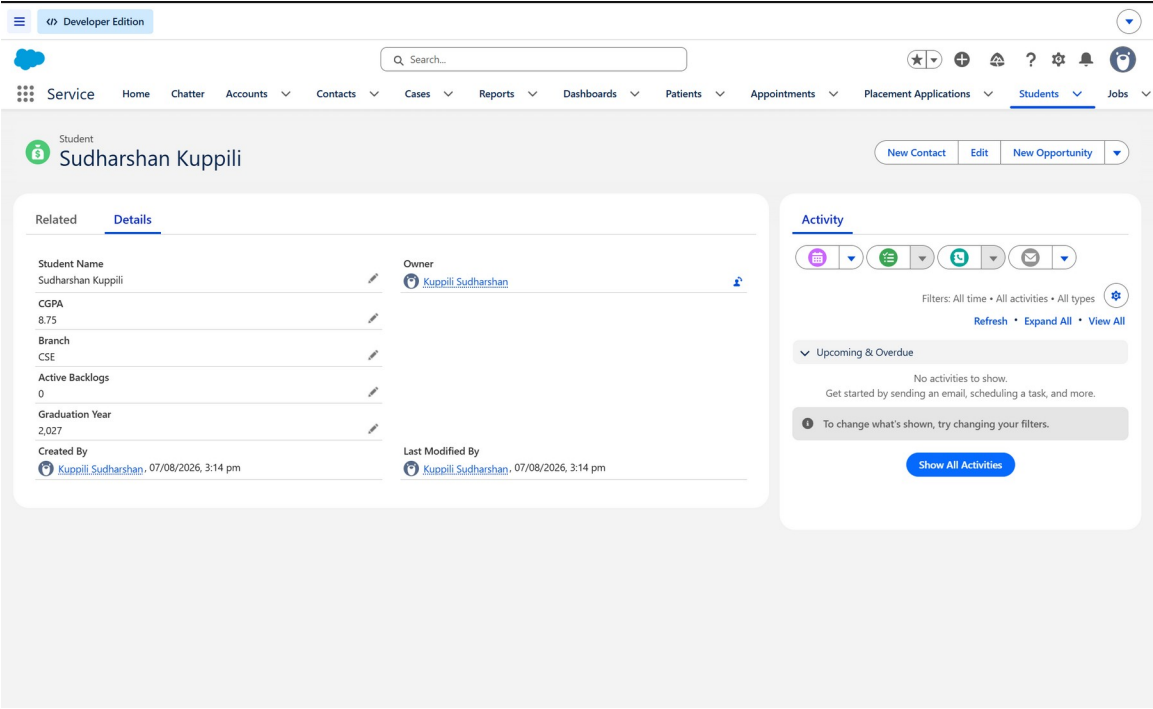


Figure 26: Student record — Sudharshan Kuppili, CGPA 8.75, Branch CSE

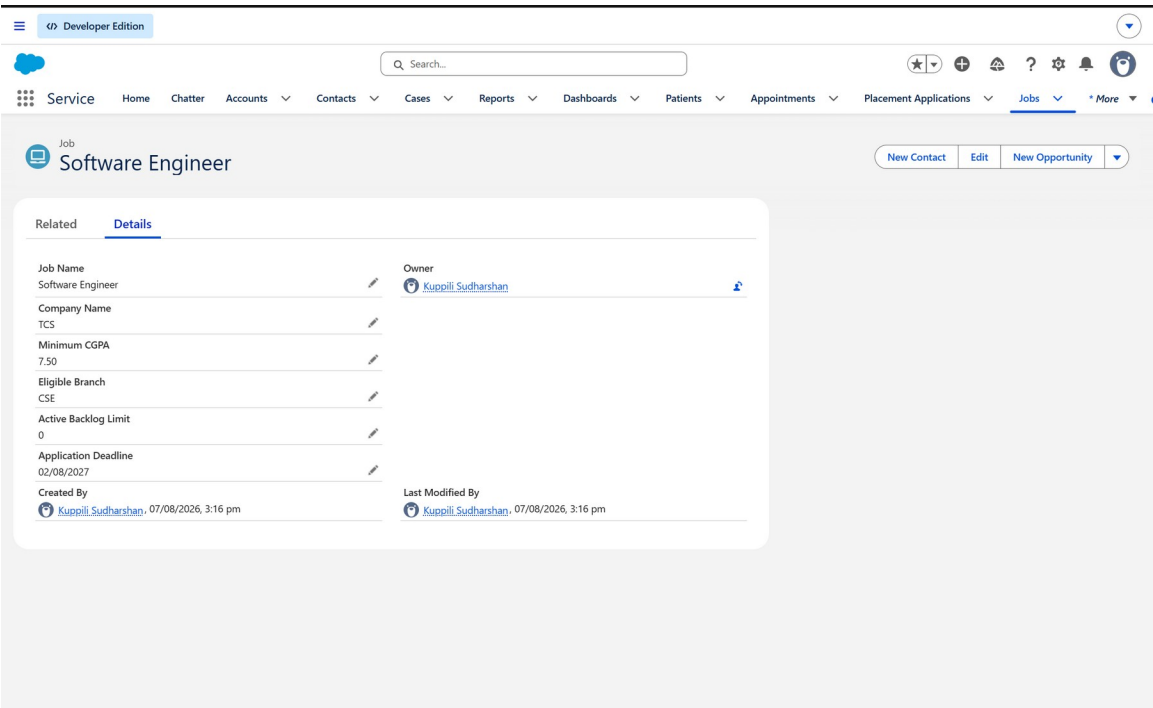


Figure 27: Job record — Software Engineer, TCS, Minimum CGPA 7.50, Application Deadline 02/08/2027

Developer Edition

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Placement Applications

Students

Jobs

Placement Application

PA-0001

New ContactEditNew Opportunity

RelatedDetails

Application Number

PA-0001

Owner

Kuppli Sudharshan

Student

Sudharshan Kuppli

Job

Software Engineer

Application Date

07/08/2026

Status

Applied

Remarks

First Placement Application

Created By

Kuppli Sudharshan, 07/08/2026, 3:23 pm

Last Modified By

Kuppli Sudharshan, 07/08/2026, 3:23 pm

Activity

Filters: All time • All activities • All types

RefreshExpand AllView All

Upcoming & Overdue

No activities to show.

Get started by sending an email, scheduling a task, and more.

To change what's shown, try changing your filters.

Show All Activities

Figure 28: Placement Application PA-0001 — Student and Job linked, Application Date auto-populated, Status: Applied